

THE \$30 DRUM THAT STOPS YOUR STOVE HEATING THE SKY

Why a barrel stove throws a third of your wood up the chimney, how a second drum catches it back, and the safety line you cannot cross.

WHAT IT IS

A barrel stove is a steel drum on legs with a cast iron door and a 6 inch flue collar. No baffle inside, no second burn for the smoke. Stoves built that way land around 40 percent efficient.

On a leaky old stove, plain hot gas can carry off about a third of everything you burn. That is the heat the second drum goes after.

You stack a second steel drum on top and route the smoke through it before it reaches the chimney. On paper, that gets you around a third more heat from the same wood.



Two stacked drums. The lower one burns, the upper one soaks heat from the smoke before it hits the chimney.

SHOPPING LIST

- ☐ **Barrel stove kit** (United States Stove Company). Cast iron door, legs, 6 inch flue collar with damper, and a flue brush. Lists about 90 dollars, sells around 70.
- ☐ **Double barrel adapter kit.** Cast iron collars and spacer legs that join the two drums. About 66 to 80 dollars.
- ☐ **Two used 55 gallon steel drums,** about 10 to 25 dollars each. Only a drum the stove maker says is suitable. Plain steel, never galvanized, never one that held fuel, solvent, or chemicals.
- ☐ **Magnetic stovepipe thermometer,** about 15 dollars. One for each end.
- ☐ **Moisture meter,** about 20 dollars.
- ☐ **Sand,** about 2 inches in the bottom barrel. The manufacturer requires it.
- ☐ **Stovepipe, hearth pad, and listed chimney parts.** Extra, and not optional.

WHAT IT COSTS VS A CERTIFIED STOVE

about \$200	The barrel stove body, all in with used drums, pipe, and a thermometer. Safe venting, floor protection, and chimney cost extra.
\$1,899	A certified steel stove like a Drolet, for the stove alone.
\$2,300 to \$4,200	A Blaze King, stove only.
\$4,500 to \$5,500	What a 3,000 dollar certified stove often becomes once it is installed.

Certified stoves are cleaner and far more efficient, and in a house they are the right answer. The drum is a shop and cabin heater, nothing more.

WHY THE DRUM CATCHES HEAT

14 times

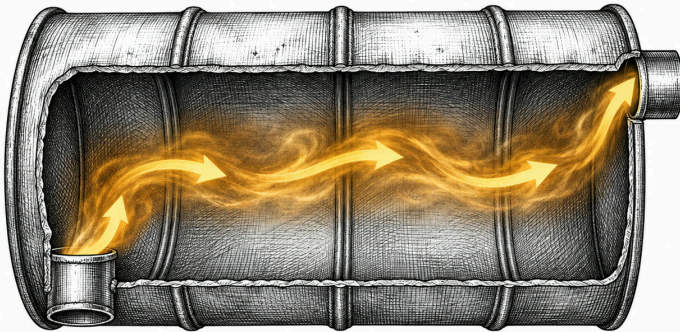
A 6 inch pipe has about 28 square inches of opening. The inside of the drum has close to 400. The gas crawls through about 14 times slower and sits against steel longer.

+22 sq ft

A 6 to 8 foot pipe run is roughly 9 to 12 square feet of steel. The drum adds about 22 square feet, roughly tripling the hot steel along that run.

4 times

A 400 degree surface throws about four times the radiant heat of a 200 degree one into a 70 degree room. Radiant heat climbs steeply with temperature.



Gas enters one end low, crosses the whole drum, and leaves the far end high. Collars must be at opposite ends.

THE WORKED EXAMPLE

Say the gas leaves the fire barrel at 700 degrees Fahrenheit and the top drum pulls it down to 450. Room air is 70.

250 caught out of a 630 gap = about 40 percent recovered

If hot gas was carrying off a third of your fuel, catching 40 percent of it is worth about 13 points of efficiency. The stove climbs from about 40 percent to the low to mid 50s. That is roughly a third more heat per log, or one cord in four you never burn.

This is a worked example, not a tested result. Only a controlled efficiency test would give you your real number.

THE BUILD, IN ORDER

- 1** Assemble the barrel stove kit on the lower drum, exactly per the manual.
- 2** Use only a drum the stove maker identifies as suitable. Plain steel, never galvanized. Never a drum that held fuel, solvent, or chemicals. Unknown residue can explode.
- 3** Put about 2 inches of sand in the bottom barrel. It keeps the coals off the thin steel floor.
- 4** Fit the adapter. One collar on top of the fire barrel, the other on the belly of the upper drum at one end. Bolt the spacer legs to both drums.
- 5** Set the two collars at opposite ends, so the gas crosses the whole drum. Close together, the far end sits cold.
- 6** Remove the lower damper and install it in the upper outlet collar. Attach your stovepipe to that collar.
- 7** Level and support the drum the way the manual shows. An empty drum weighs 40 to 55 pounds.
- 8** Cure any paint or liner outdoors first, away from anything that can burn. Do the first fire outside.

READ THE MANUAL FIRST

The kit is not a suggestion. Follow the maker's assembly, the sand bed, the damper move, and the collar positions exactly. Improvised placement changes how it drafts and how it burns.

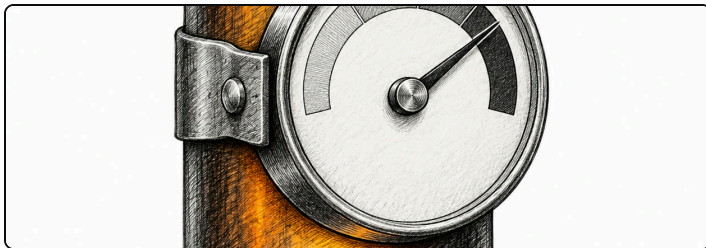
THE CREOSOTE CATCH

Wood smoke carries tar and water vapor. Cool it too far and it condenses on the steel as creosote, a crust that burns. The National Fire Protection Association ties failure to clean, mostly creosote, to about 1 in 4 home heating fires.

That warning applies to this build too. A barrel stove starts with hotter exhaust, so there is room to take heat out, but only if you watch the temperature.

under 250	Surface reading on the pipe. The creosote zone. Burn hotter with dry wood, smaller splits, and more air.
250 to 475	You are burning right.
above 475	Overfiring. Close the air down. You are wasting fuel and cooking the steel.

Put a thermometer on the pipe above the top drum and above the fire barrel, and watch both ends. Dry wood means under 20 percent moisture. Split a log and test the fresh face.



A magnetic thermometer on single wall pipe is your best clue to how cool the smoke runs. Follow the maker's placement.

CLEANING THE TOP DRUM

- ☐ Let the stove go completely cold first.
- ☐ Inspect the connector and chimney at least twice a month in heating season, and the whole upper drum, not just one collar view.
- ☐ Creosote can collect in the upper drum, the connector, and the chimney. Check all of them.
- ☐ A thin, dry, powdery coat brushes out. If it looks shiny or glazed, stop using the stove and call a chimney professional.

SAFETY RULES. NOT OPTIONAL.

- ☐ **Plain steel drum only.** Never galvanized. Burning zinc throws fumes that make you sick. Never a drum that held fuel, solvent, or chemicals.
- ☐ **Not certified, not for a home.** This kit is sold for shops, garages, cabins, and camp. It cannot be sold or used in Washington, Oregon, or California.
- ☐ **Clearances.** An unlisted stove needs 36 inches to anything that can burn, on every side and above. Single wall pipe needs 18 inches. Approved shielding with an air gap can reduce those.
- ☐ **Floor.** Set it on a noncombustible hearth pad that extends past the stove on every side. On a wood floor, the pad is not optional.
- ☐ **Insurance.** Homemade and uncertified stoves usually are not covered. Ask your insurer before you light one.
- ☐ **The steel wears out.** A drum is 18 to 20 gauge, thinner than a coat hanger. The fire barrel can burn through after a few hard seasons. A new drum is a cheap swap.
- ☐ **Draft.** Cooling the gas weakens the chimney's pull. If smoke enters the room, stop and get the venting checked. Do not just assume more chimney height.
- ☐ **Chimney.** Rise at least 3 feet above the roof, and 2 feet above anything within 10 feet. Use a listed insulated chimney section and thimble through any wall or roof.
- ☐ **Never leave it running with nobody around.** Put a carbon monoxide alarm in the space. If it sounds, get everyone outside and call emergency services.

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